

Tanay Pratap Singh

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EDUCATION

Master's of Science in Applied Data Science
B.Tech in Computer Science and Engineering

Syracuse University, Syracuse, NY
JSSATE Noida, UP, India

Jan. 2025 – Dec. 2026
Dec. 2020 – May 2024

EXPERIENCE

Research Assistant

Feb. 2026 – Present

Syracuse University

Syracuse, NY

- Developed data ingestion workflow with Python, Pandas, and Zooniverse API to parse 276,000+ classifications, reconciling 5 schemas across 25,104 glitch subjects and 8,293 channels.
- Designed image processing workflow using OpenCV to retrieve 49,000 spectrograms via batch API calls, transforming 1200x1200 mosaic images into paired 224x224 channel heatmaps for deep learning input.
- Trained two-input Convolutional Neural Network with shared MobileNetV2 backbone in TensorFlow/Keras to predict volunteer similarity labels from paired spectrograms; achieved validation AUC of 0.89 on 2,000-subject holdout set.
- Reduced Expected Calibration Error from 0.10 to 0.05 using label smoothing, AdamW optimizer, and AUC-based early stopping; produced glitch-type by auxiliary-channel similarity matrix to map detector subsystems to glitch morphologies.

Data and Product Analyst Intern

Jul. 2023 – Aug. 2023

ProProfs

Noida, UP, India

- Performed product and customer analytics using Python and Excel, delivering SWOT, competitive benchmarking (100+ features vs. Zoho and Salesforce), and ICP analyses across SaaS products serving 5,000+ users.
- Analyzed customer datasets of 2,000 clients using Python and Excel pivot tables to identify behavior patterns and conversion drivers, enabling the product team to prioritize high-impact features that increased trial-to-paid conversion.
- Created interactive Tableau dashboards from SQL and Excel data to visualize feature usage, customer segmentation, and conversion funnels, helping stakeholders identify under-used features and reduce churn by 5 percent.

PROJECTS

ZTF Transient Classifier

May 2026 – Jun. 2026

- * Engineered end-to-end Python streaming pipeline ingesting real-time ZTF astronomical alerts via Apache Kafka and Avro, processing 27,000 nightly records and decoding FITS images with fastavro and Astropy for ML models.
- * Prepared robust analytics engineering layer in dbt on cloud data warehouse, developing 8 SQL models across layers with advanced techniques, ensuring data integrity.
- * Built multimodal deep learning classifier combining CNN for image cutouts and tabular feature branch, served through FastAPI REST API.
- * Orchestrated nightly ETL workflow with Apache Airflow and tracked model drift using Population Stability Index.
- * Established CI/CD pipeline in GitHub Actions with Docker containerization; data warehouse portable across DuckDB, Google BigQuery, and Snowflake.

ShopFloor AI

Jan. 2026 – Feb. 2026

- * Architected predictive maintenance system using Python, scikit-learn, and MLflow on 10,000 manufacturing sensor records with 5 failure modes, engineering 16 features including power output, thermal differential, and overstrain metrics; best model (Gradient Boosting) achieved 0.85 F1 score and 0.96 AUC-ROC.
- * Implemented real-time streaming architecture using Apache Kafka for sensor data, computed rolling KPIs with programmatic anomaly detection, and integrated RAG using LangChain, ChromaDB, and OpenAI API to enable natural-language search over manufacturing documentation.
- * Deployed 7-service containerized application with Docker Compose including FastAPI, Streamlit, Apache Airflow with automated daily retraining and data quality gates, and MLflow for model registry promotion.

TECHNICAL SKILLS

Languages and Frameworks: Python, SQL, R, JavaScript, HTML/CSS, FastAPI, REST API, Streamlit

Machine Learning and AI: Scikit-learn, TensorFlow, Keras, XGBoost, CatBoost, LangChain, OpenCV, SHAP, LLM, Generative AI, RAG, Random Forest, Gradient Boosting, SVM, LSTM, CNN, Convolutional Neural Network, NLP, Deep Learning, Ensemble Methods, Classification, Predictive Modeling, Feature Engineering, Cross-Validation, Model Deployment

Data and Analytics: Pandas, NumPy, Spark, Tableau, Power BI, DAX, Power Query, Excel, Plotly, Matplotlib; EDA, Data Visualization, Data Wrangling, A/B Testing, Time Series Forecasting, Hypothesis Testing, Anomaly Detection, Cohort Analysis, Funnel Analysis, Customer Segmentation

MLOps, Cloud and Data Engineering: Docker, Apache Kafka, Apache Airflow, dbt, MLflow, Model Registry, DuckDB, Snowflake, BigQuery, MySQL, Avro, ChromaDB, Vector Databases, ETL, ELT, Data Pipelines, Streaming Data, Data Warehousing, Data Modeling, Dimensional Modeling, Analytics Engineering, Data Quality Testing, CI/CD, GitHub Actions, Git, Azure, GCP, Jupyter

Statistics, Modeling and Business: Linear Regression, Logistic Regression, ANOVA, PCA, Lean Six Sigma, DMAIC, SIPOC, Pugh Matrix, Process Mapping, Process Improvement, KPI Design, Business Intelligence, Stakeholder Communication